Tadiwa Mandava

Computer Engineering · University of Oklahoma

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Education

University of Oklahoma Expected May 2028

B.S. Computer Engineering, Minor in Mathematics · Norman, OK

Experience

Software Engineering Intern January 2026 → Present

K20 Center, University of Oklahoma · Norman, OK

- Develop and deploy front-end and back-end web applications and Unity/C# educational games used by K–12 teachers and students; test across Center projects, reproducing and filing defects
- Built and shipped Hive, a classroom platform for curating and publishing boards of links, images, and video; owned the stack from PostgreSQL schema through REST API to deployed React SPA

IT Service Delivery Specialist September 2024 → Present

University of Oklahoma IT · Norman, OK

- Administer identity and access management and Microsoft 365 services across 30,000+ user accounts, automating account provisioning and recurring administrative workflows with PowerShell
- Operate on a five-person cross-functional team handling ticket-driven telecom, account, and backend operations requests, sustaining a 98.4% satisfaction rate

Projects

LSM Storage Engine

Rust · tokio · proptest · criterion

- Built a crash-safe, concurrent key-value storage engine with an LSM-tree architecture: write-ahead log, immutable SSTables, Bloom filters, and background compaction, exposing Redis-compatible commands
- Validated compaction, WAL recovery, and concurrency correctness with property-based tests against a reference HashMap

LLM Inference Engine

`C++ · Python · AVX2 · OpenMP · Emscripten`

- Implemented GPT-2 inference from scratch in C++, including tokenization, transformer blocks, multi-head attention, KV caching, sampling, and tensor ops, without an ML inference framework
- Optimized CPU inference with cache-aware matmul, SIMD, and multithreading; implemented INT8/INT4 quantization, validated against a Hugging Face reference

CUDA Matrix Multiplication

`CUDA C++ · Nsight Compute · PyTorch`

- Implemented an SGEMM optimization ladder (coalesced loads, shared-memory tiling, register blocking, thread coarsening, vectorized loads), validating each kernel against cuBLAS on non-power-of-two shapes before benchmarking
- Profiled each stage in Nsight Compute using achieved occupancy, DRAM and SM throughput, warp stall reasons, and register pressure, selecting each optimization from the binding hardware constraint rather than intuition
- Implemented a fused FlashAttention kernel that tiles Q/K/V through the GPU memory hierarchy instead of materializing the full attention matrix, exposed as a PyTorch custom operator verified against SDPA

Hive, a Classroom Curation Platform

`TypeScript · JavaScript · SQL · React · AdonisJS · PostgreSQL · Vite`

- Built and shipped Hive, a classroom platform for curating and publishing boards of links, images, and video; owned the stack from PostgreSQL schema through REST API to deployed React SPA

Bare-Metal RTOS on STM32

`C · ARM assembly · STM32F446 · arm-none-eabi-gcc · OpenOCD · GDB`

Skills

Languages	Python · ARM assembly · C · C++ · CUDA C++ · JavaScript · Rust · SQL · SystemVerilog · TypeScript
Frameworks & tools	AdonisJS · arm-none-eabi-gcc · AVX2 · Basys 3 · criterion · Emscripten · GDB · GTKWave · Nsight Compute · OpenMP · OpenOCD · PostgreSQL · proptest · PyTorch · React · STM32F446 · tokio · Verilator · Vite · Vivado